

Longitudinal Analysis of Multimodal Pain Management Prescribing Among Spine Surgeons

United States, 2014–2023
Medicare Part B & Part D Claims Data

Data Source: Centers for Medicare & Medicaid Services (CMS)
Medicare Provider Utilization and Payment Data

Key Findings

Study Population: 6,678 unique procedure-confirmed spine surgeons identified across 2014–2023 (neurosurgery + orthopedic surgery performing spine CPT codes). ~4,200 spine surgeons per year in 2014 declining to ~3,925 in 2023.

Opioid Prescribing (↓51.7%): Total opioid claims fell from 707,906 (2014) to 341,801 (2023). Claims per provider declined from 190.3 to 135.6 (–28.7%). Average days supply per opioid claim decreased from 15.4 to 11.1 days (–28.3%). Total opioid drug cost dropped from \$18.8M to \$4.4M (–76.8%).

Gabapentinoid Prescribing (↑39.3%): Total claims rose from 151,485 to 211,067. Claims per provider increased from 59.7 to 86.4 (+44.7%).

Muscle Relaxant Prescribing (↑59.3%): Total claims rose from 113,116 to 180,212. Claims per provider increased from 58.2 to 81.6 (+40.3%).

NSAID Prescribing (↑22.8%): Total claims rose from 135,554 to 166,415. Claims per provider increased from 81.6 to 103.4 (+26.7%).

Proportion Shift: Opioids comprised 63.9% of all pain management claims in 2014 but only 38.0% by 2023. Non-opioid modalities (gabapentinoids, muscle relaxants, NSAIDs) collectively grew from 36.1% to 62.0%.

Table 1: Total Claims by Drug Category and Year

Year	Opioid	Gabapentinoid	Muscle Relaxant	NSAID
2014	707,906	151,485	113,116	135,554
2015	632,887	162,994	106,785	140,799
2016	620,368	179,237	104,936	145,562
2018	511,154	192,547	127,850	145,705
2019	449,991	187,295	137,559	150,059
2022	360,937	213,107	165,541	157,433
2023	341,801	211,067	180,212	166,415

Note: 2017, 2020, and 2021 data unavailable due to CMS API/format changes.

Table 2: Mean Claims Per Provider by Drug Category and Year

Year	Opioid	Gabapentinoid	Muscle Relaxant	NSAID
2014	190.3	59.7	58.2	81.6
2015	174.1	63.4	55.4	82.3
2016	170.4	67.7	57.2	82.4
2018	150.9	71.8	60.8	86.1
2019	139.4	71.0	65.0	88.1
2022	135.3	85.4	76.4	98.8
2023	135.6	86.4	81.6	103.4

Table 3: Average Days Supply Per Claim

Year	Opioid	Gabapentinoid	Muscle Relaxant	NSAID
2014	15.4	33.7	23.4	32.9
2015	16.5	34.0	24.1	33.5
2016	16.6	34.3	24.6	33.9
2018	15.0	35.0	24.3	36.0
2019	12.8	35.6	23.6	36.3
2022	11.4	34.1	22.4	35.2
2023	11.1	34.2	22.3	35.2

Figure 1: Total Pain Management Claims by Drug Category

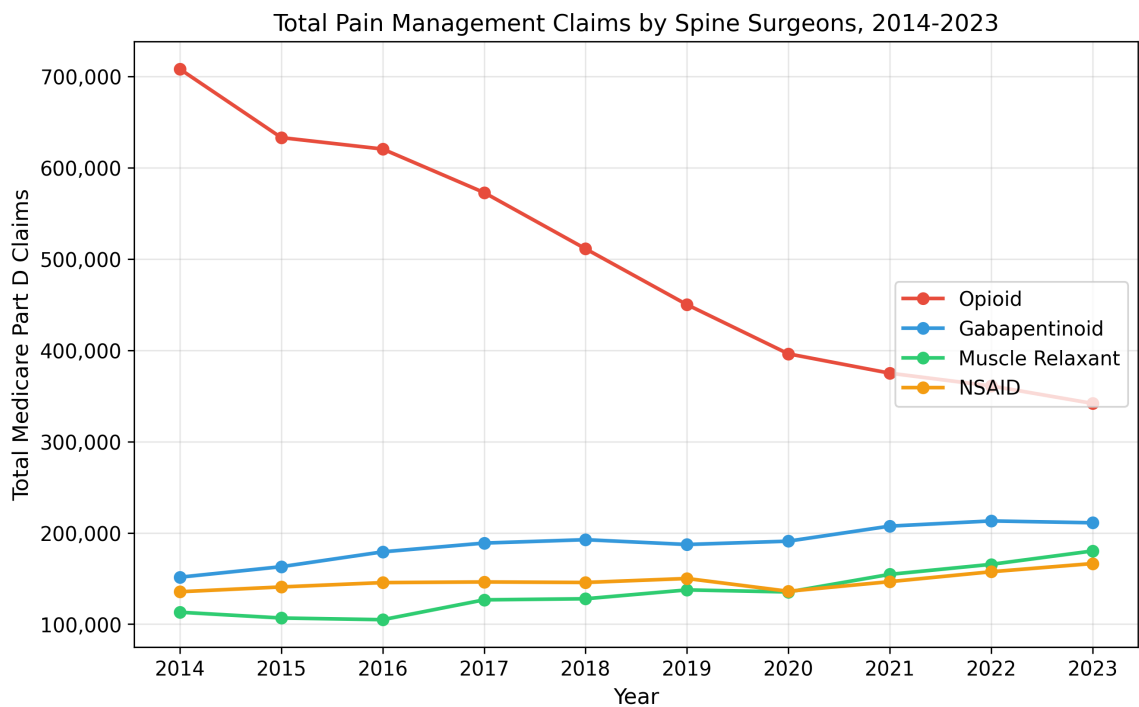


Figure 2: Mean Claims Per Spine Surgeon Prescriber

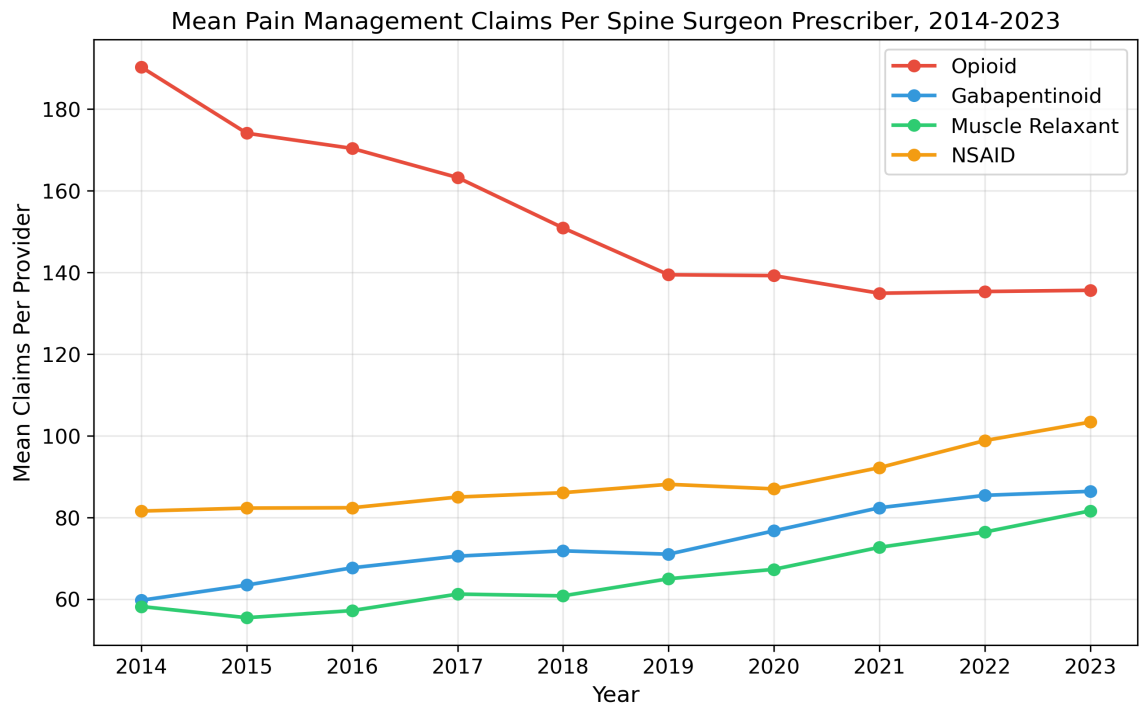


Figure 3: Number of Spine Surgeon Prescribers by Category

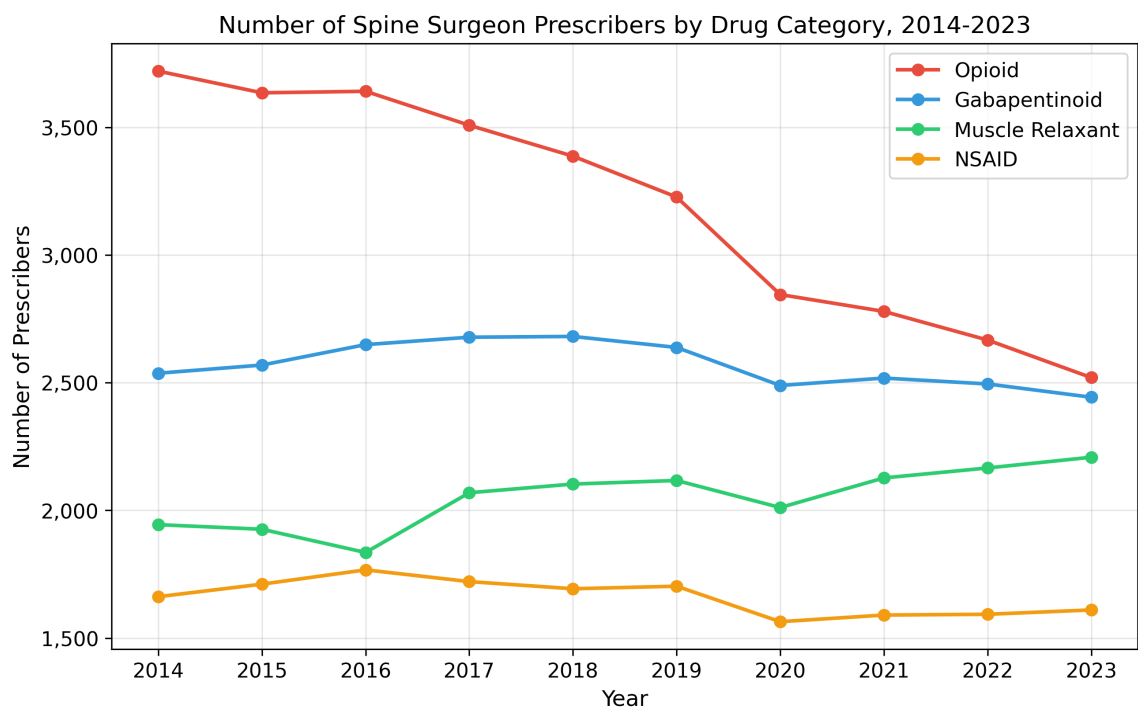


Figure 4: Average Prescription Duration (Days Supply Per Claim)

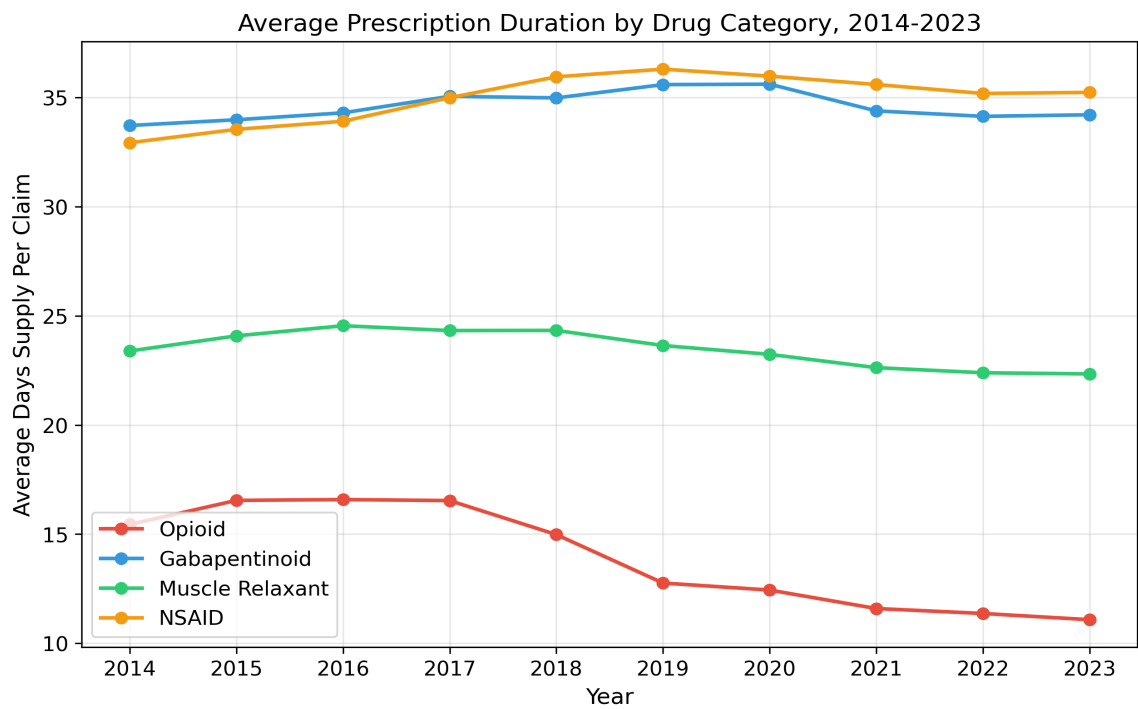


Figure 5: Normalized Prescribing Trends (2014 = 100)

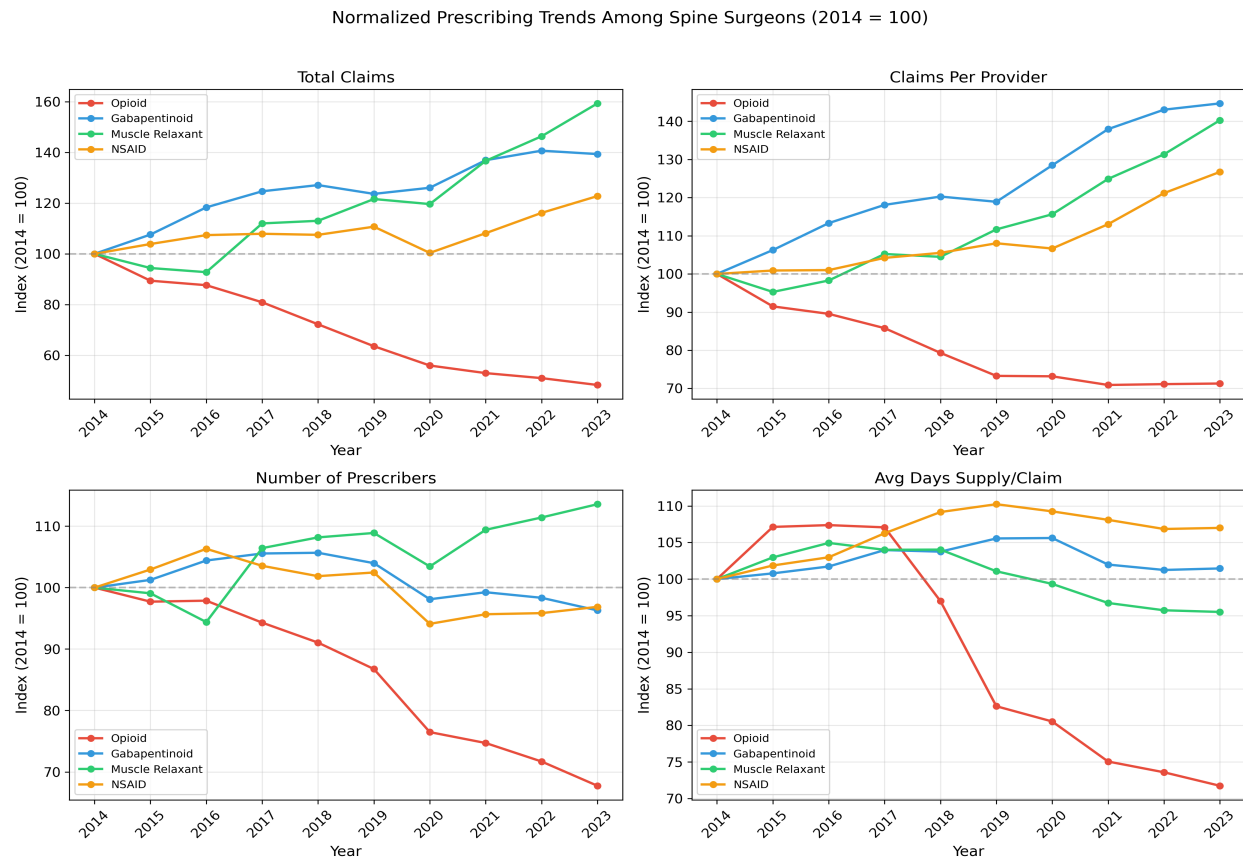


Figure 6: Total Drug Cost by Category

Total Drug Cost by Category Among Spine Surgeons, 2014-2023

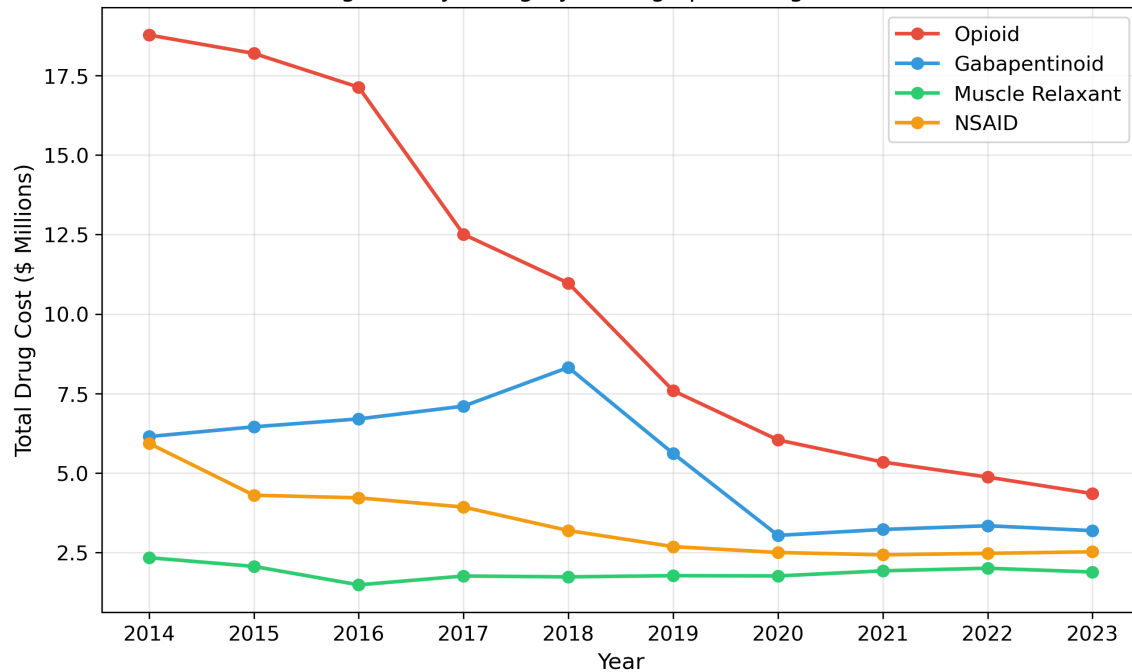


Figure 7: Distribution of Pain Management Claims by Category

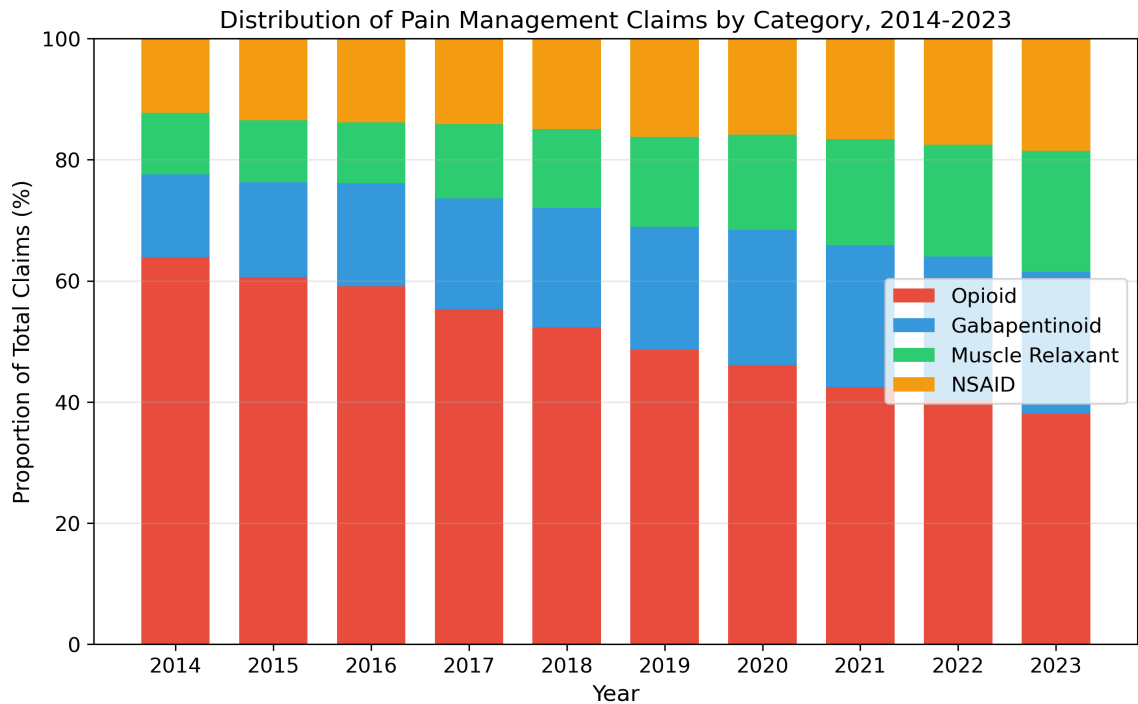
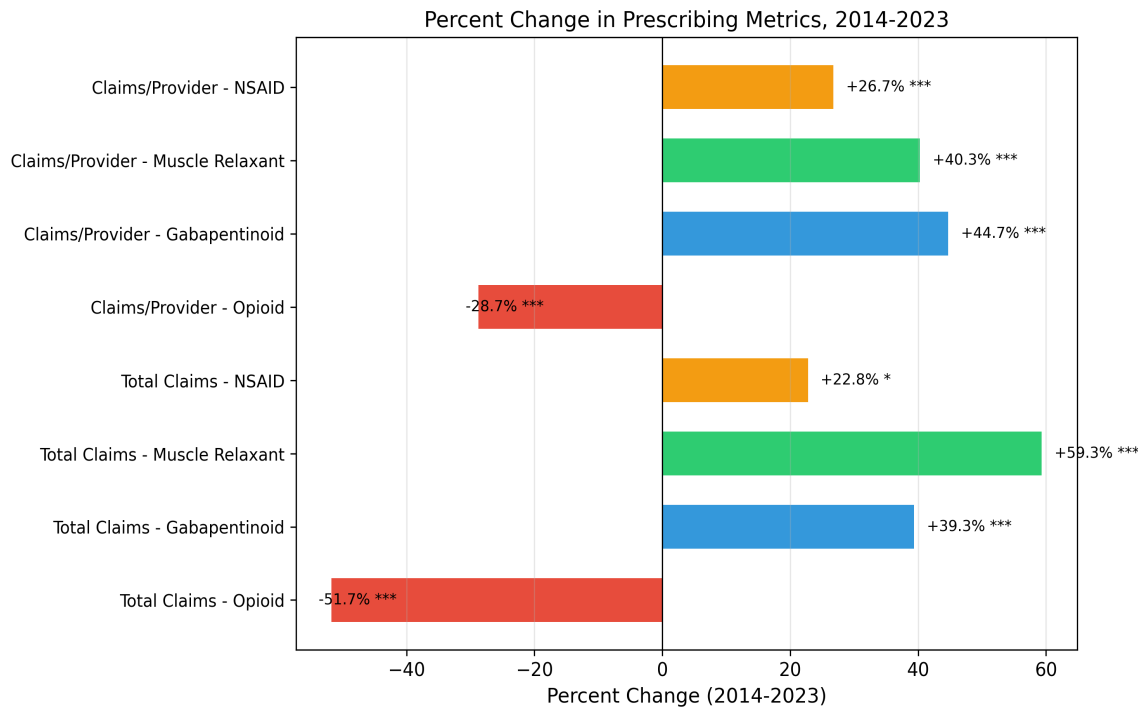


Figure 8: Percent Change in Prescribing Metrics, 2014–2023



Methods

Data Source: CMS Medicare Provider Utilization and Payment Data, accessed via the CMS data.cms.gov API (Socrata Open Data API).

Spine Surgeon Identification: Providers were identified from Medicare Part B Physician/Supplier data as those with provider type 'Neurological Surgery' or 'Orthopedic Surgery' who billed for spine-specific CPT codes (22551, 22554, 22612, 22630, 22633, 22840, 22842, 63005, 63030, 63042, 63047, 63056, 63075, 63081, 22800, 22802, 22804, 22808, 22810, 22812, among others). This procedure-confirmed approach ensures the cohort represents surgeons actively performing spine procedures.

Drug Categories: Four categories of pain management medications were analyzed: (1) **Opioids** — hydrocodone, oxycodone, tramadol, morphine, fentanyl, codeine, hydromorphone, methadone, oxymorphone, tapentadol, buprenorphine, meperidine; (2) **Gabapentinoids** — gabapentin, pregabalin; (3) **Muscle Relaxants** — cyclobenzaprine, methocarbamol, tizanidine, baclofen, metaxalone, carisoprodol, orphenadrine, dantrolene, chlorzoxazone; (4) **NSAIDs** — meloxicam, diclofenac, naproxen, celecoxib, ibuprofen, indomethacin, ketorolac, piroxicam, sulindac, etodolac, nabumetone, ketoprofen, oxaprozin, fenoprofen, flurbiprofen, mefenamic acid, tolmetin.

Prescribing Metrics: Total Medicare Part D claims, number of unique prescribers, mean claims per provider, average days supply per claim, total drug cost, and total 30-day fill equivalents were calculated for each drug category per year.

Statistical Analysis: Linear regression was used to estimate temporal trends (slope per year) for each metric. Percent change was calculated from 2014 baseline to 2023. Years with missing data (2017, 2020, 2021) were excluded from regression.

Limitations: Data for 2017, 2020, and 2021 were unavailable due to CMS API access issues or changes in data format. The analysis is limited to Medicare Part D claims and does not capture prescriptions filled through other payers. Spine surgeon identification is based on Medicare billing and may not capture all spine surgeons.